

# Appendix S1 – Supplementary Information

Beaver engineering creates intense but bidirectional dissolved organic carbon hotspots in streams

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This document contains the material supporting the main findings of the manuscript (the hypothesised causal structure and tested hypotheses of the structural causal model, priors, residuals, posterior predictive checks, posterior parameter distributions, a stream-level classification of  $\Delta\text{DOC}$ , the total effects of the drivers on  $\Delta\text{DOC}$ , and an alternative SCM including lateral stream–wetland connectivity). The full description of data collection, variable derivation and model specification is given in the Methods of the main manuscript.

## SCM: hypothesised causal structure

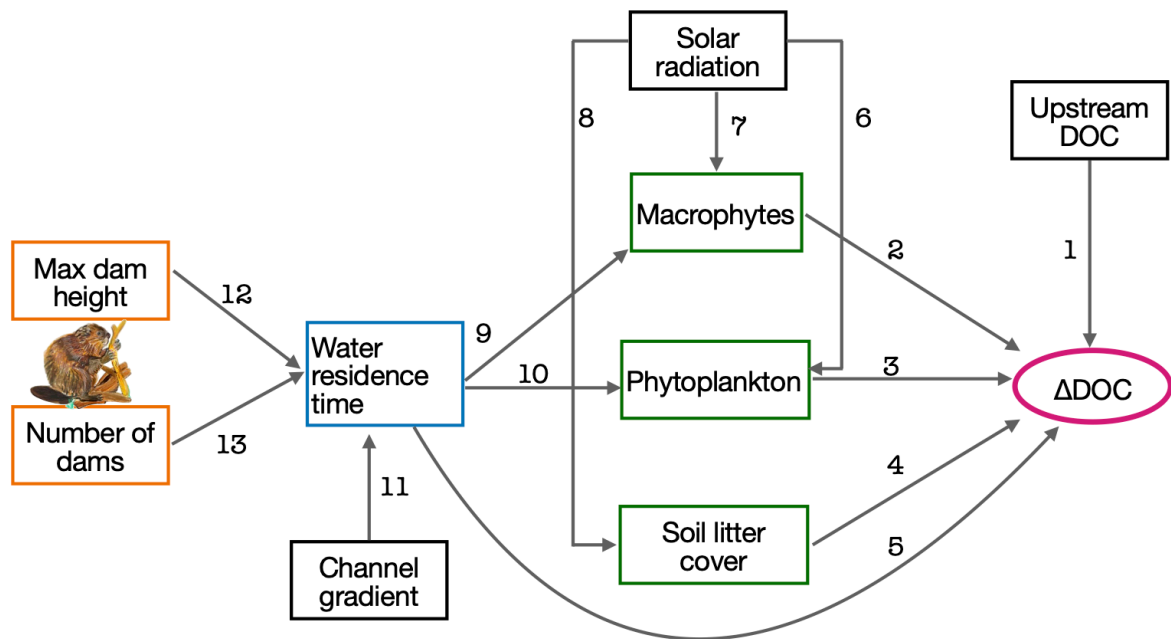


Figure S1: Directed acyclic graph (DAG) of the structural causal model (SCM), with the hypothesised causal mechanisms numbered 1–13. Orange boxes: beaver-engineering variables; the blue box: stream hydrology; green boxes: primary producer variables; black boxes: parent drivers not influenced by other variables; the magenta ellipse: the outcome  $\Delta\text{DOC}$ . Each numbered edge corresponds to one tested hypothesis in Table S1. Beaver illustration credits to © Sarah Edmonds Illustration.

## SCM: tested hypothesis

Table S1: Tested hypotheses for the structural causal model (SCM). The causal-path numbers correspond to the numbered edges of the DAG in Figure S1 (and Figure 1b of the main document).

| Causal path | Predictor    | Response           | Hypothesis                                                                                                                                                                                                                                                                                                                                        |
|-------------|--------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1           | DOC upstream | $\Delta\text{DOC}$ | If DOC upstream increases, then $\Delta\text{DOC}$ decreases because beaver ponds can act as sink, retaining or transforming DOC through processes such as microbial degradation, sedimentation, and photodegradation, thereby reducing downstream DOC concentrations relative to upstream inputs (Larson et al. 2007; Wohl 2013; Johnston 2014). |
| 2           | Macrophytes  | $\Delta\text{DOC}$ | If macrophyte abundance increases, we expect $\Delta\text{DOC}$ to increase because the decomposition of macrophyte biomass releases labile dissolved organic carbon, enhancing downstream concentrations (Reitsema et al. 2018).                                                                                                                 |

*continued on next page*

Table S1 (continued)

| Causal path | Predictor                        | Response             | Hypothesis                                                                                                                                                                                                                                                                                                                   |
|-------------|----------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3           | Phytoplankton                    | $\Delta DOC$         | If phytoplankton abundance increases, then $\Delta DOC$ increases is likely to increase, because phytoplankton contribute to DOC content through exudation during growth and release upon cell lysis, thereby enhancing downstream DOC levels relative to upstream inputs (Søndergaard et al. 1985; Sell and Overbeck 1992). |
| 4           | Soil litter cover                | $\Delta DOC$         | If soil litter cover increases, then $\Delta DOC$ increases as decomposition of litter releases dissolved organic compounds into the water (Aitkenhead et al. 1999; Cincotta et al. 2019).                                                                                                                                   |
| 5           | Water residence time             | $\Delta DOC$         | If water residence time increases, then $\Delta DOC$ increases, because longer water residence times allow for more extensive microbial decomposition and processing of dissolved organic carbon, enhancing downstream DOC concentrations (Jepsen et al. 2019; Abbasi et al. 2024).                                          |
| 6           | Solar radiation                  | Phytoplankton        | If solar radiation increases then phytoplankton abundance is expected to increase because greater light availability enhances photosynthetic rates (Reynolds 2006).                                                                                                                                                          |
| 7           | Solar radiation                  | Macrophytes          | If solar radiation increases, then macrophyte abundance increases because higher light availability enhances photosynthetic activity and supports plant growth (Feuchtmayr et al. 2009).                                                                                                                                     |
| 8           | Solar radiation                  | Litter cover         | If solar radiation increases, as in summer, then soil litter cover decreases because lower rates of tree leaf fall coincide with enhanced decomposition driven by higher temperatures and solar radiation (Gessner et al. 1999).                                                                                             |
| 9           | Water residence time             | Macrophytes          | If water residence time increases, macrophyte abundance may increase because longer residence time allows for higher nutrient retention, promoting plant growth and increasing the likelihood of macrophyte establishment and persistence (Manolaki et al. 2021).                                                            |
| 10          | Water residence time             | Phytoplankton        | If water residence time increases, phytoplankton abundance may increase because longer residence time allows for greater nutrient retention and stabilization, which can support enhanced phytoplankton growth, particularly in systems with nutrient limitation (Finger et al. 2007).                                       |
| 11          | Channel gradient (terrain slope) | Water residence time | If valley slope increases, then water residence time decrease because steeper terrain causes water to move more quickly through the landscape, reducing the time available for subsurface and surface interactions (McGuire et al. 2005).                                                                                    |

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Table S1 (continued)

| Causal path | Predictor      | Response             | Hypothesis                                                                                                                                                                                                                                                     |
|-------------|----------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12          | Max dam height | Water residence time | If dam height increases, water residence time increases because taller dams impound more water, expanding ponded areas and slowing streamflow, which prolongs the time water remains within the stream system (Majerova et al. 2015; Dittbrenner et al. 2022). |
| 13          | Number of dams | Water residence time | If the number of beaver dams increases, water residence time is expected to increase because dams slow down streamflow and create ponded areas that retain water for longer periods (Westbrook et al. 2006; Nyssen et al. 2011; Majerova et al. 2015).         |

## Structural Causal Model: the priors

To ensure the model was not over-constrained by prior knowledge, we used weakly informative priors for all parameters. This approach allows the data to inform the model without assuming strong beliefs about the parameter values beforehand (Figure S2). We note, however, that for macrophytes, phytoplankton, and litter cover—variables measured in 16 of the 176 streams—we used more informative priors that incorporate established ecological knowledge about their influence on DOC concentrations (Figure S2 panel b).

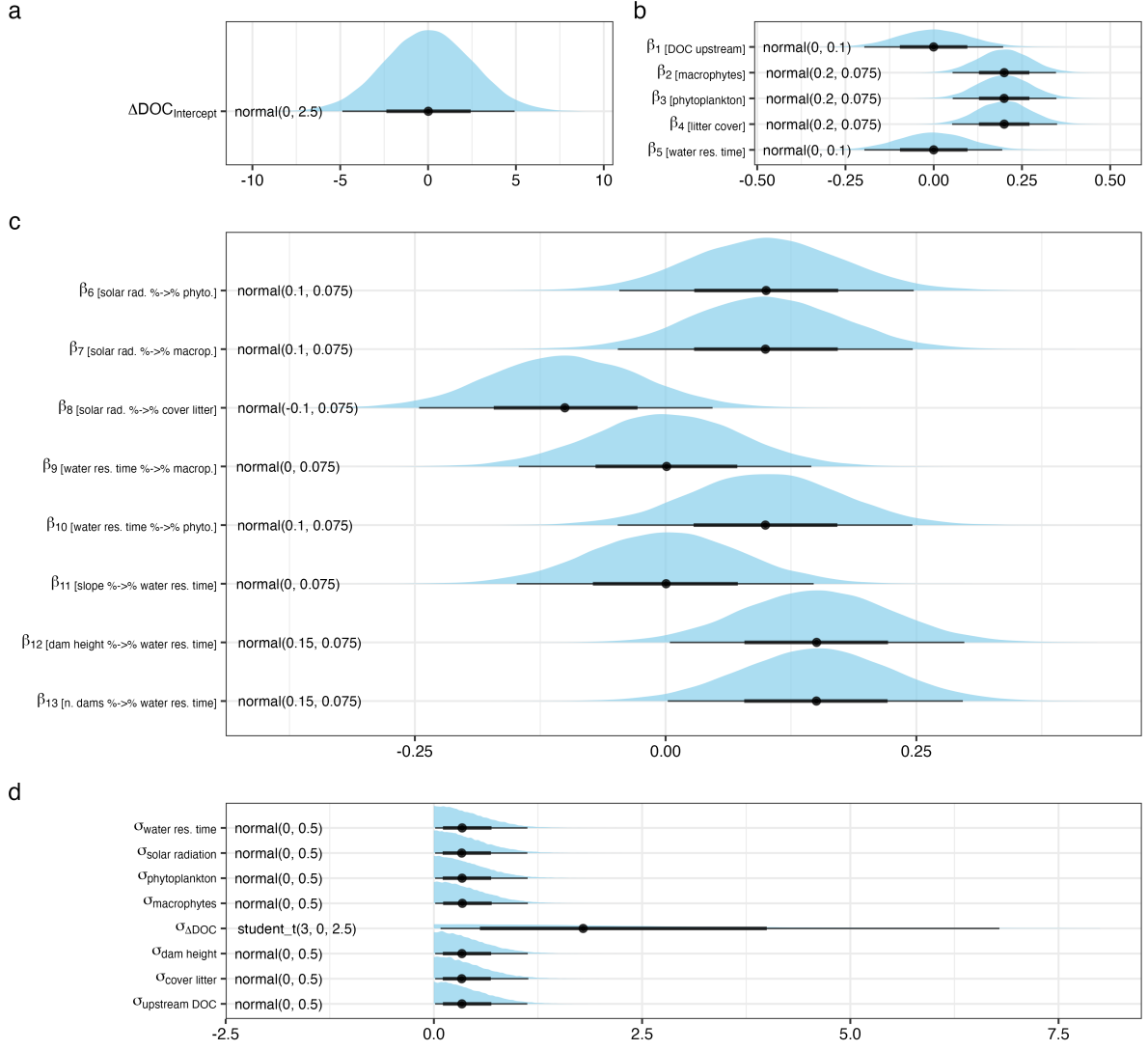


Figure S2: Half-eye plots display the prior distributions used in the model. The shaded areas (slabs) represent the probability density of each prior distribution, illustrating their shape and spread. The central dot indicates the median of the distribution, while the thick and thin horizontal lines represent the 66% and 95% credible intervals, respectively. Text labels to the left of each distribution indicate the corresponding prior specification.

## Structural Causal Model: the residuals

We can evaluate the model fit by looking at the difference between observed data and their predictions. That are the residuals ( $r_i$ ).

$$r_i = y_i - y_i^{\text{pred}} \quad (\text{S1})$$

If the model is a reasonable representation of the observational process that originates the data then the residuals should look roughly randomly scattered in comparison to a horizontal line.

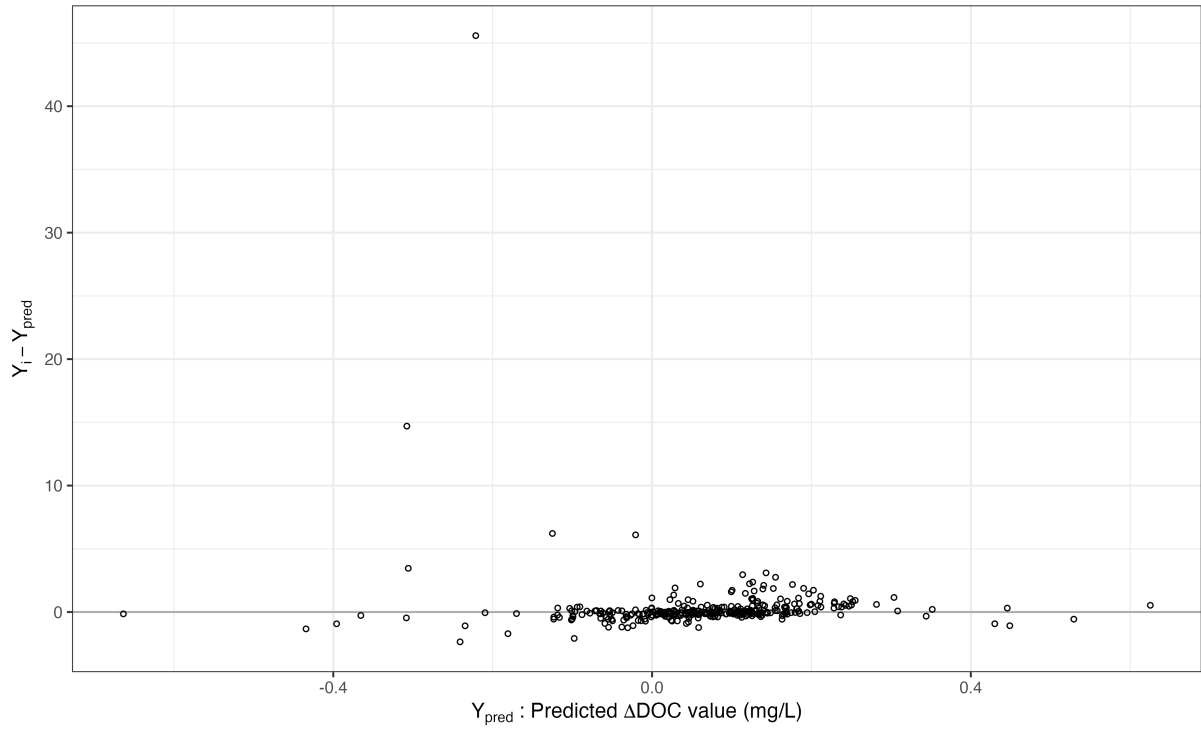


Figure S3: Residual plot showing the relationship between model-predicted  $\Delta\text{DOC}$  values ( $\text{mg L}^{-1}$ ) and corresponding residuals. Points represent the difference between observed and predicted values; a random scatter around zero (grey line) indicates good model fit and no systematic bias in predictions.

## Structural Causal Model: posterior predictive checks

Posterior predictive checks assess how well the fitted SCM model reproduces observed data by comparing the observed values to data simulated from the model's posterior predictive distribution.

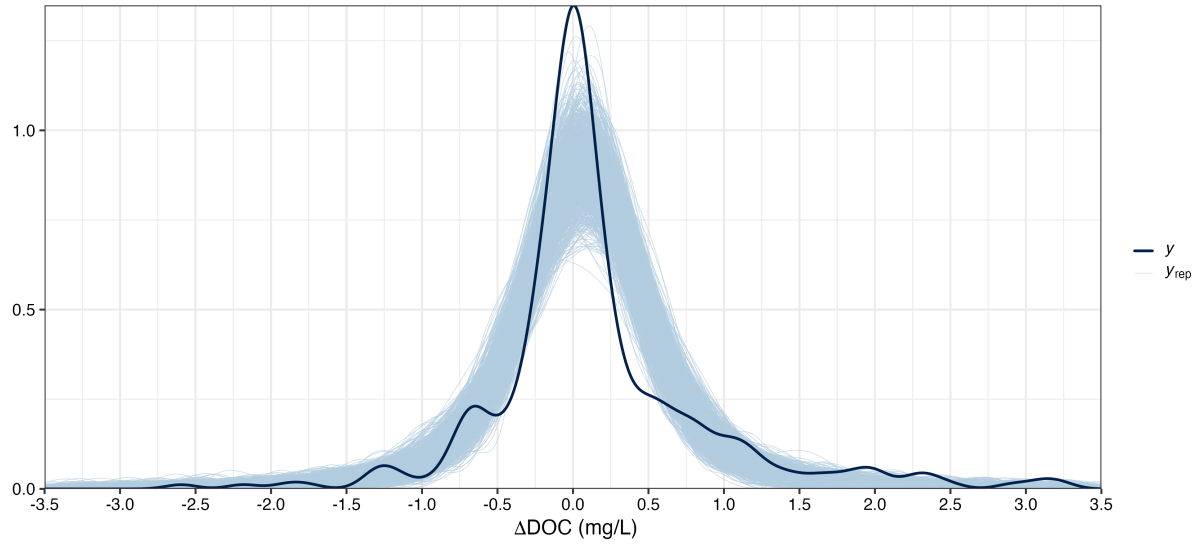


Figure S4: Posterior predictive checks of the mean and standard deviation statistics for the response variable delta DOC, with 1000 posterior draws used to generate the predictive distribution. This plot provides a visual assessment of the model's fit by comparing the observed data (dark blue) against the model's predictions (light blue) from the posterior distribution.

## Structural Causal Model: posterior parameters distribution

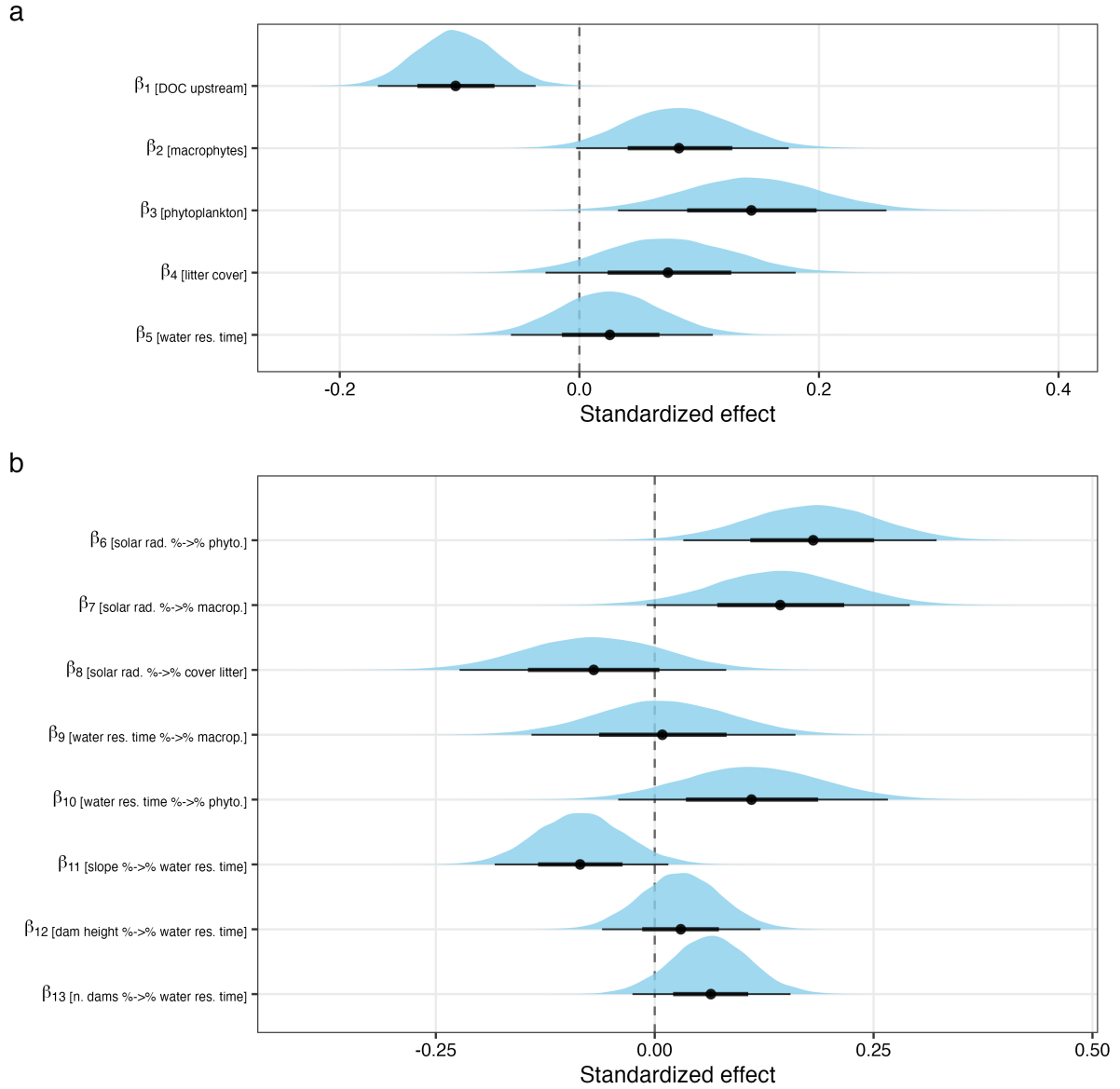


Figure S5: Posterior parameters distributions of the fitted SCM. Median shown with point; 66% and 95% credible intervals shown with black bars. Parameters that have more than 90% probability mass on either side of zero are considered to have a positive or negative influence on the response variable. a) The direct effects on  $\Delta\text{DOC}$  b) The indirect effects on  $\Delta\text{DOC}$ .

## Structural Causal Model: stream-level classification of $\Delta\text{DOC}$

The large variability of individual responses shown in Figure 1a (main document) can arise in two fundamentally different ways: either streams differ consistently from each other – some being stable DOC sources, others stable sinks – or each stream fluctuates between source, negligible and sink behaviour from one sampling occasion to the next. Distinguishing these two possibilities matters for interpretation and management: only in the first case could one map and manage “source streams” and “sink streams”; the sink/negligible/source percentages of Figure 1a are probabilities per sampling and must not be read as percentages of streams. To answer this question we classified each stream individually using the structural causal model (SCM), whose stream-level random intercept captures any consistent between-stream differences. (As a

consistency check, the posterior predictive distribution of  $\Delta\text{DOC}$  from the SCM for a typical stream – predictors at their mean, stream effects integrated out – gives 12% sink, 76% negligible and 12% source responses, in close agreement with the season-specific measurement-error model of Figure 1a.) For each upstream–downstream unit (a few of the 176 streams comprised more than one separately sampled beaver-dam sequence) we computed the posterior distribution of the expected  $\Delta\text{DOC}$  (population-level effects plus the stream-level intercept, averaged over the stream’s observations; 2,000 posterior draws) and assigned each unit to the region (sink  $< -0.5$   $\text{mg L}^{-1}$ , ROPE  $\pm 0.5$   $\text{mg L}^{-1}$ , source  $> 0.5$   $\text{mg L}^{-1}$ ) that held the majority of its posterior mass. The expected  $\Delta\text{DOC}$  of every unit fell within the ROPE (posterior medians from  $-0.41$  to  $0.38$   $\text{mg L}^{-1}$ ; median width of the 90% credible interval  $0.42$   $\text{mg L}^{-1}$ ; Figure S6), reflecting the small average effect (intercept  $0.00$   $\text{mg L}^{-1}$ ; 95% BCI  $-0.12$ ;  $0.12$ ) and the small between-stream standard deviation ( $0.05$   $\text{mg L}^{-1}$ ; 95% BCI  $0.00$ ;  $0.14$ ). The answer is therefore the second possibility: no stream shows consistent source or sink behaviour, and the heavy-tailed posterior predictive distribution of Figure 1a captures the variability of individual samplings around a null average rather than stable stream-level differences.

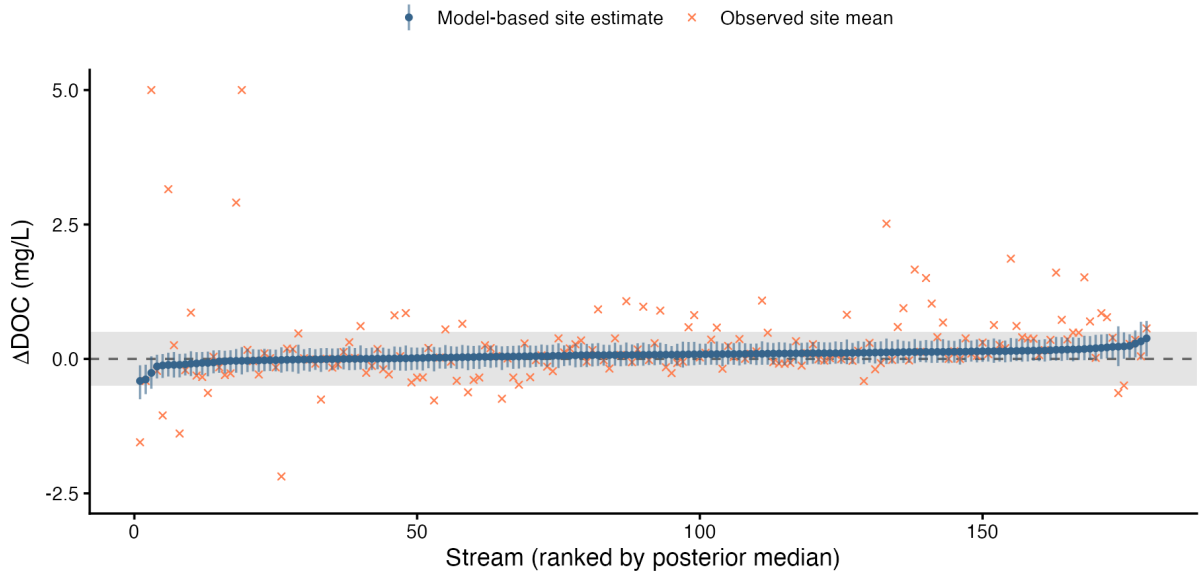


Figure S6: Stream-level  $\Delta\text{DOC}$ . Blue points and bars: posterior median and 90% credible interval of the model-based expected  $\Delta\text{DOC}$  of each stream (ranked by posterior median). Orange crosses: observed stream-averaged  $\Delta\text{DOC}$ , shown for reference (one value of  $29.9$   $\text{mg L}^{-1}$  is clipped at  $5$   $\text{mg L}^{-1}$ ). The grey band is the region of practical equivalence ( $\pm 0.5$   $\text{mg L}^{-1}$ ).

## Structural Causal Model: total effects of the exogenous drivers on $\Delta\text{DOC}$

The edges of the SCM (Figure 1b, main document) are direct, path-specific effects. To summarize the overall influence of each exogenous driver on  $\Delta\text{DOC}$  we computed its *total effect* as the sum of the products of the coefficients along every open causal path from the driver to  $\Delta\text{DOC}$  (e.g., for the number of dams: dams  $\rightarrow$  WRT  $\rightarrow$   $\Delta\text{DOC}$ , dams  $\rightarrow$  WRT  $\rightarrow$  macrophytes  $\rightarrow$   $\Delta\text{DOC}$ , and dams  $\rightarrow$  WRT  $\rightarrow$  phytoplankton  $\rightarrow$   $\Delta\text{DOC}$ ), evaluated on each posterior draw (path tracing; Hayes 2018). All effects are on the standardized scale (change in  $\Delta\text{DOC}$ , in standard deviations, per standard deviation of the driver).

Upstream DOC concentration had the largest total effect ( $-0.10$ ; 90% BCI:  $-0.16$ ;  $-0.05$ ;  $P(\beta < 0) > 0.99$ ), followed by solar radiation, which acts through all three primary-producer mediators ( $+0.04$ ; 90% BCI:  $0.01$ ;  $0.07$ ;  $P(\beta > 0) = 0.99$ ). The total effect of water residence time ( $+0.04$ ; 90% BCI:  $-0.02$ ;  $0.11$ ;  $P(\beta > 0) = 0.87$ ) was positive but less certain. The total effects of the beaver-engineering variables, which act only through water residence time, were small: number of dams  $+0.003$  (90% BCI:  $-0.002$ ;  $0.010$ ;  $P(\beta > 0) = 0.82$ ) and maximum dam height  $+0.001$  (90% BCI:  $-0.003$ ;  $0.007$ ;  $P(\beta > 0) = 0.68$ ); channel gradient had a small negative total effect ( $-0.004$ ; 90% BCI:  $-0.012$ ;  $0.002$ ). Hence, the direction of the beaver-mediated DOC change is set primarily by the external context of the pond (incoming DOC load and light availability for primary producers), whereas the structural characteristics of the dam complex modulate it only weakly via water residence time (Figure S7).

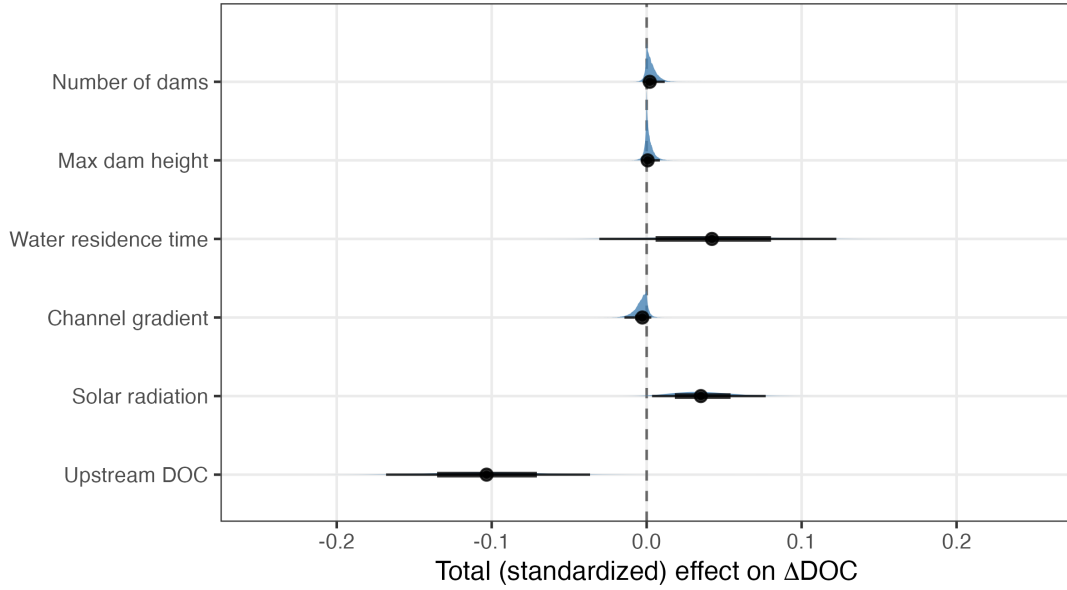


Figure S7: Posterior distributions of the total (standardized) effect of each exogenous driver on  $\Delta\text{DOC}$ , obtained by tracing all open causal paths of the SCM. Points: posterior means; thick and thin bars: 66% and 95% credible intervals.

## Alternative structural causal model: lateral stream–wetland connectivity

Because a DAG encodes a specific set of causal assumptions, we also tested an alternative, extended SCM that adds one causal path to the main SCM (Figure S1): path 14, from the lateral connectivity between the stream and adjacent wetlands to water residence time (WRT; Figure S8). Our hypothesised mechanism is that a hydrologically connected wetland acts as an off-channel transient-storage zone: water moves laterally into and out of this storage compartment, extending the time it resides in the reach before continuing downstream (transient-storage theory; Bencala and Walters 1983; Wohl 2021). We therefore hypothesised that beaver-pond WRT increases with the degree of lateral stream–wetland connectivity and, consequently, that connectivity indirectly influences  $\Delta\text{DOC}$  via WRT and the primary producers.

**Connectivity variable.** For each site, we first determined whether a wetland was present within a 50 m buffer around the river reach between the upstream and downstream sampling locations, using the wetland layer (marshes, bogs, swamp forests, wet meadows, floodplains

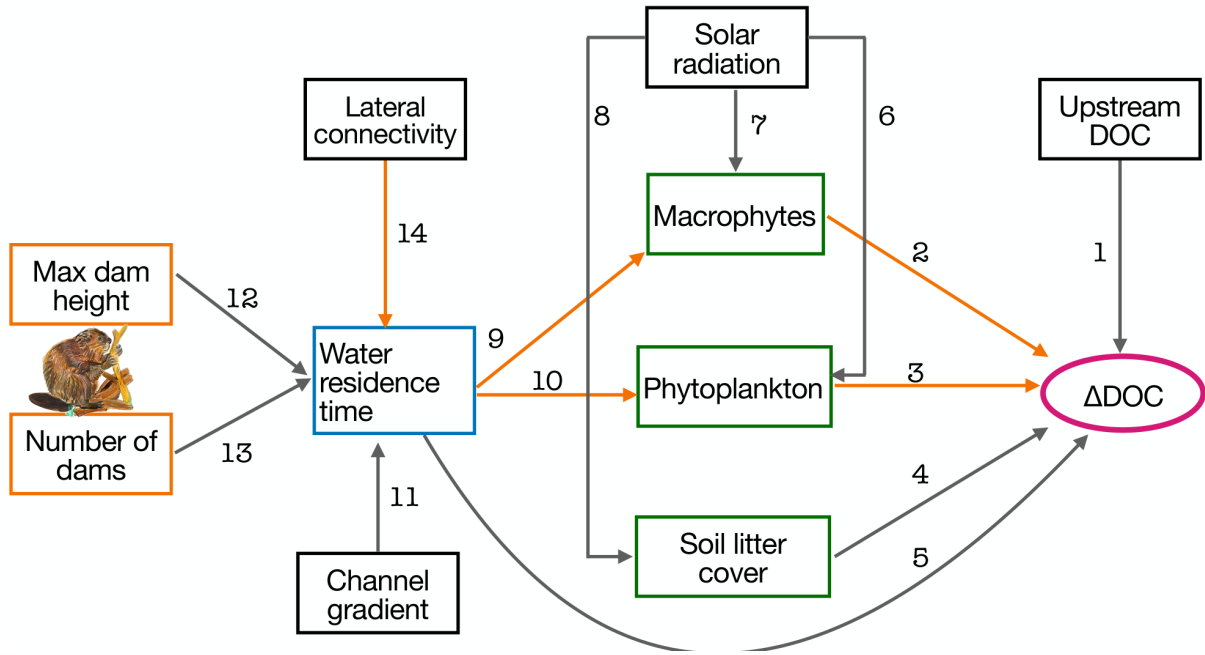


Figure S8: Extended directed acyclic graph (DAG) of the alternative SCM. It is identical to the main SCM (Figure S1) except for the additional causal path 14, connecting lateral stream–wetland connectivity to water residence time. Orange edges mark the open front-door paths through which the total causal effect of lateral connectivity flows to  $\Delta\text{DOC}$  (paths 14  $\rightarrow$  9  $\rightarrow$  2, 14  $\rightarrow$  10  $\rightarrow$  3, and 14  $\rightarrow$  5).

and fens) of the swissTLM3D topographic landscape model (SwissTopo 2024). For sites with a wetland, we then assessed visually on the swisstopo maps and orthophotos whether the wetland was hydrologically connected to the river water, or separated from it by roads, trails, tree lines or other barriers. Each site was thereby assigned to one of three categories: *No wetland* (no wetland within the buffer;  $n = 136$  sites), *Wetland, not connected* (wetland present but not directly connected to the river;  $n = 11$ ), and *Wetland, connected* (wetland present and directly connected to the river;  $n = 13$ ). Twenty-two sites could not be classified and were excluded from this analysis. Note that most of these wetlands are themselves strongly influenced by beaver damming and would likely not exist without beavers.

**Identification: why the full SCM does not need to be re-fitted.** We encoded the extended DAG with the *dagitty* R package (function `dagitty()`) and queried it with `adjustmentSets(dag, exposure = "lateral_connectivity", outcome = "delta_DOC", effect = "total")`, which returned the empty set, and `paths(dag, "lateral_connectivity", "delta_DOC")`, which lists the open paths (orange in Figure S8). The empty adjustment set follows from the structure of the DAG: lateral connectivity has no parents (it is exogenous), so no back-door (confounding) path into  $\Delta\text{DOC}$  exists, and its entire causal influence flows forward through its single child, WRT. For the same reason the total effect of WRT on  $\Delta\text{DOC}$  is identified without adjustment. Consequently, the estimand “total effect of connectivity on  $\Delta\text{DOC}$ ” equals the product of two unconfounded quantities – the effect of connectivity on WRT and the total effect of WRT on  $\Delta\text{DOC}$  – and a parsimonious model containing only the WRT equation and a  $\Delta\text{DOC}$ -on-WRT equation identifies it exactly. Re-fitting the entire eight-response SCM would leave this estimand unchanged (the mediator equations for macrophytes and phytoplankton are not needed when the mediated effect is captured through the total  $\text{WRT} \rightarrow \Delta\text{DOC}$  path), while adding substantial computational cost and imputation machinery for variables that are irrelevant to the query.

We therefore fitted the following two-equation Bayesian model (plus an intercept-only equation to impute missing dam heights) in *brms*:

**Outcome:**  $\Delta\text{DOC}_i \mid \text{mi}(0.283) \sim \text{Student-}t(\nu, \mu_i, \sigma_{\Delta\text{DOC}}), \quad \mu_i = \alpha + \beta_5 \cdot \text{WRT}_i + u_{\text{site}[i]}$

**Mediator:**  $\text{WRT}_i \mid \text{mi}() \sim \mathcal{N}(\eta_i, \sigma_{\text{WRT}})$

$$\eta_i = \beta_{13}\text{NDams}_i + \beta_{12}\text{DamHeight}_i + \beta_{11}\text{Slope}_i + \gamma_{c[i]} + \beta_s\text{Solar}_i$$

**Priors:**  $\alpha \sim \mathcal{N}(0, 2.5), \quad \beta_5 \sim \mathcal{N}(0, 0.1), \quad u_{\text{site}} \sim \mathcal{N}(0, \sigma_{\text{site}}), \quad \sigma_{\text{site}} \sim \mathcal{N}(0, 2)$

$$\beta_{12}, \beta_{13} \sim \mathcal{N}(0.15, 0.075), \quad \beta_{11}, \beta_s \sim \mathcal{N}(0, 0.075), \quad \gamma_c \sim \mathcal{N}(0, 1)$$

$$\sigma_{\Delta\text{DOC}} \sim \text{Student-}t(3, 0, 2.5), \quad \sigma_{\text{WRT}} \sim \mathcal{N}(0, 0.5), \quad \nu \sim \text{Gamma}(2, 0.1) \quad (\text{S2})$$

where  $\gamma_c$  are the three connectivity-level coefficients (*No wetland* as reference for the contrasts) and the shared parameters carry the same informative priors as in the main SCM (Equation 4, main document); the measurement error of  $\Delta\text{DOC}$  ( $0.283 \text{ mg L}^{-1}$ ) and the imputation of missing WRT and dam-height values ( $\text{mi}()$ ) are handled as in the main SCM. Four chains were run ( $\hat{R} \leq 1.005$ ; 310 observations). The total effect of each connectivity level on  $\Delta\text{DOC}$  was computed on each posterior draw as the product of the connectivity  $\rightarrow$  WRT contrast and the WRT  $\rightarrow$   $\Delta\text{DOC}$  coefficient (path tracing, as for Figure S7).

**Results.** Relative to sites without wetland, the presence of an unconnected wetland increased WRT by 0.23 standardized units (90% BCI:  $-0.15; 0.61$ ;  $P(\beta > 0) = 0.84$ ) and a connected wetland by 0.00 (90% BCI:  $-0.36; 0.36$ ). The WRT  $\rightarrow$   $\Delta\text{DOC}$  path was positive but weak (0.07; 90% BCI:  $0.00; 0.14$ ). The resulting total effects of connectivity on  $\Delta\text{DOC}$  were negligible: 0.02 (90% BCI:  $-0.01; 0.06$ ) for unconnected and 0.00 (90% BCI:  $-0.03; 0.03$ ) for connected wetlands (Figure S9). We therefore found no evidence that lateral stream–wetland connectivity modifies the beaver-mediated DOC change through the water-residence-time pathway. Given that only 7% and 6% of sites fell into the connected and unconnected wetland categories, we regard this result as power-limited rather than as strong evidence of absence of an effect. Importantly, the estimates of the remaining paths were consistent with those of the main SCM (Figure 1b, main manuscript), indicating that our main conclusions are robust to this alternative causal structure.

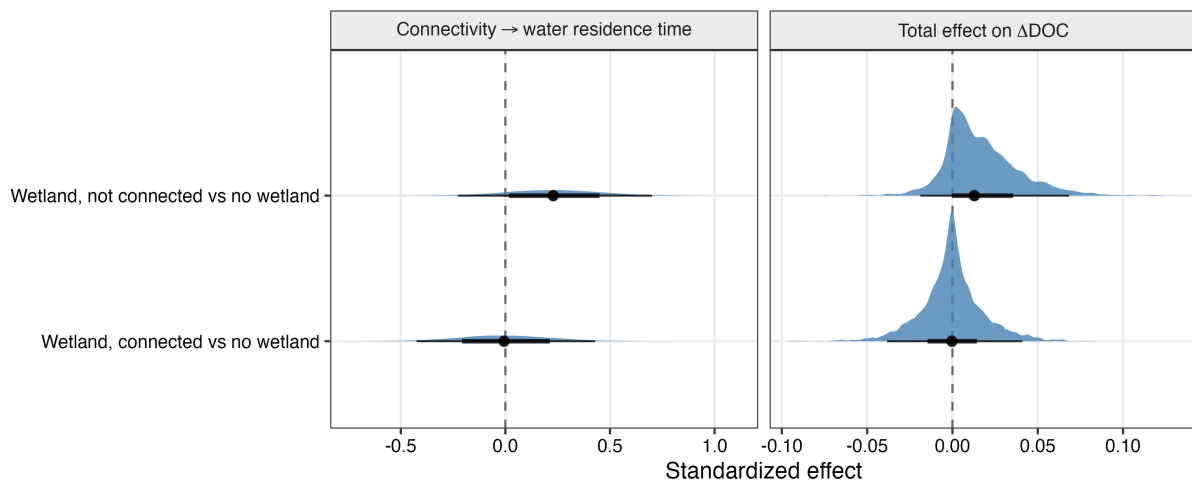


Figure S9: Posterior effects of lateral stream–wetland connectivity (relative to sites without wetland) on water residence time (left) and the total mediated effect on  $\Delta\text{DOC}$  (right), controlling for beaver-dam variables, channel gradient and solar radiation. Points: posterior medians; thick and thin bars: 66% and 95% credible intervals. The dashed line marks zero.

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